MANAGING USERS AND GROUPS

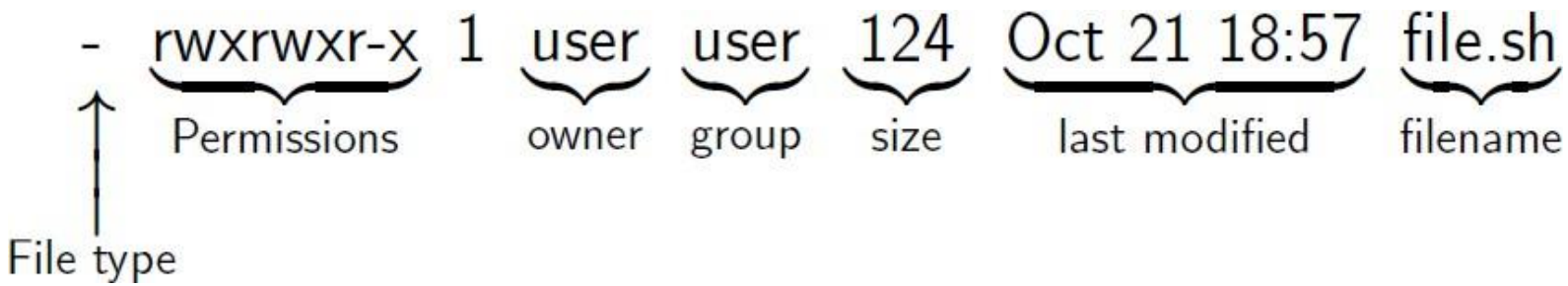
Plan

- **User and Group Concepts**
- **Changing SuperUseer Access**
- **Managing Local User Accounts**
- **Managing Local Group Accounts**

USER AND GROUP CONCEPTS

File properties

- Recapitulate ls –l
- displays information



WHAT IS A USER?

- *user* account is used to provide security boundaries between different people and programs that can run commands.

- Users have:
  - ***user names*** to identify them to human users and make them easier to work with.
  - ***user ID* or *UID*** : the unique identification number used by the system to distinguishe user accounts

User Accounts Types

- ***Superuser*** account is for administration of the system.The name of the superuser is root and the account has UID 0.The superuser has full access to the system.

- ***System user*** accounts which are used by processes that provide supporting services.These processes, or *daemons*, usually do not need to run as the superuser.They are assiged non-privileged accounts that allow them to secure their files and other resources from each other and from regular users on the system. Users do not interactively log in using a system user account.

- ***Regular user*** accounts which they use for their day-to-day work. Like system users, regular users have limited access to the system.

Display User Information

You can use the **id** command to show information about the currently logged-in user.

```
[user01@host ~]$ id
uid=1000(user01) gid=1000(user01) groups=1000(user01)
 context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
```

To view basic information about another user, pass the username to the **id** command as an argument.

```
[user01@host]$ id user02
uid=1002(user02) gid=1001(user02) groups=1001(user02)
 context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
```

Users configuration file

- The output of the preceding command displays users by name, but internally the operating system uses the **UIDs** to track users.

- The mapping of usernames to UIDs is defined in **databases of account information**. By default, systems use the **/etc/passwd** file to store information about local users.

```
1 user01: 2 x: 3 1000: 4 1000: 5 User One: 6 /home/user01: 7 /bin/bash
```

- UIDs below 1000 are reserved for system use and should not be assigned to users.
- GIDs below 1000 are reserved for system use and should not be assigned to users.

root@server:~

File Edit View Search Terminal Help

```
[root@server ~]# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
```

For brevity several entries are omitted

```
sanjay:x:1001:1001::/home/sanjay:/bin/bash
rpmtester:x:1002:1002::/home/rpmtester:/bin/bash
a:x:1003:1003::/home/a:/bin/bash
b:x:1004:1004::/home/b:/bin/bash
c:x:1005:1007::/home/c:/bin/bash
d:x:1006:1007::/home/d:/bin/bash
test-owner:x:1007:1009::/home/test-owner:/bin/bash
test-group:x:1008:1010::/home/test-group:/bin/bash
test-other:x:1009:1012::/home/test-other:/bin/bash
mamu:x:1010:1013::/home/mamu:/bin/bash
billa:x:1011:1014::/home/billa:/bin/bash
regularuser:x:1012:1015::/home/regularuser:/bin/bash
maya:x:1013:1016::/home/maya:/bin/bash
mimi:x:1014:1017::/home/mimi:/bin/bash
mini:x:1015:1018::/home/mini:/bin/bash
habu:x:1016:1020::/home/habu:/bin/bash
rhuser:x:1017:1021::/home/rhuser:/bin/bash
mamua:x:1018:1022::/home/mamua:/bin/bash
mamama:x:1019:1023::/home/mamama:/bin/bash
[root@server ~]#
```

WHAT IS A GROUP?

- A group is a collection of users that need to share access to files and other system resources.

- the system distinguishes groups by the unique identification number the *group ID* or *GID*.

- The mapping of group names to GIDs is defined in databases of group account information /etc/group

```
1 group01: 2 x: 3 10000: 4 user01,user02,user03
```

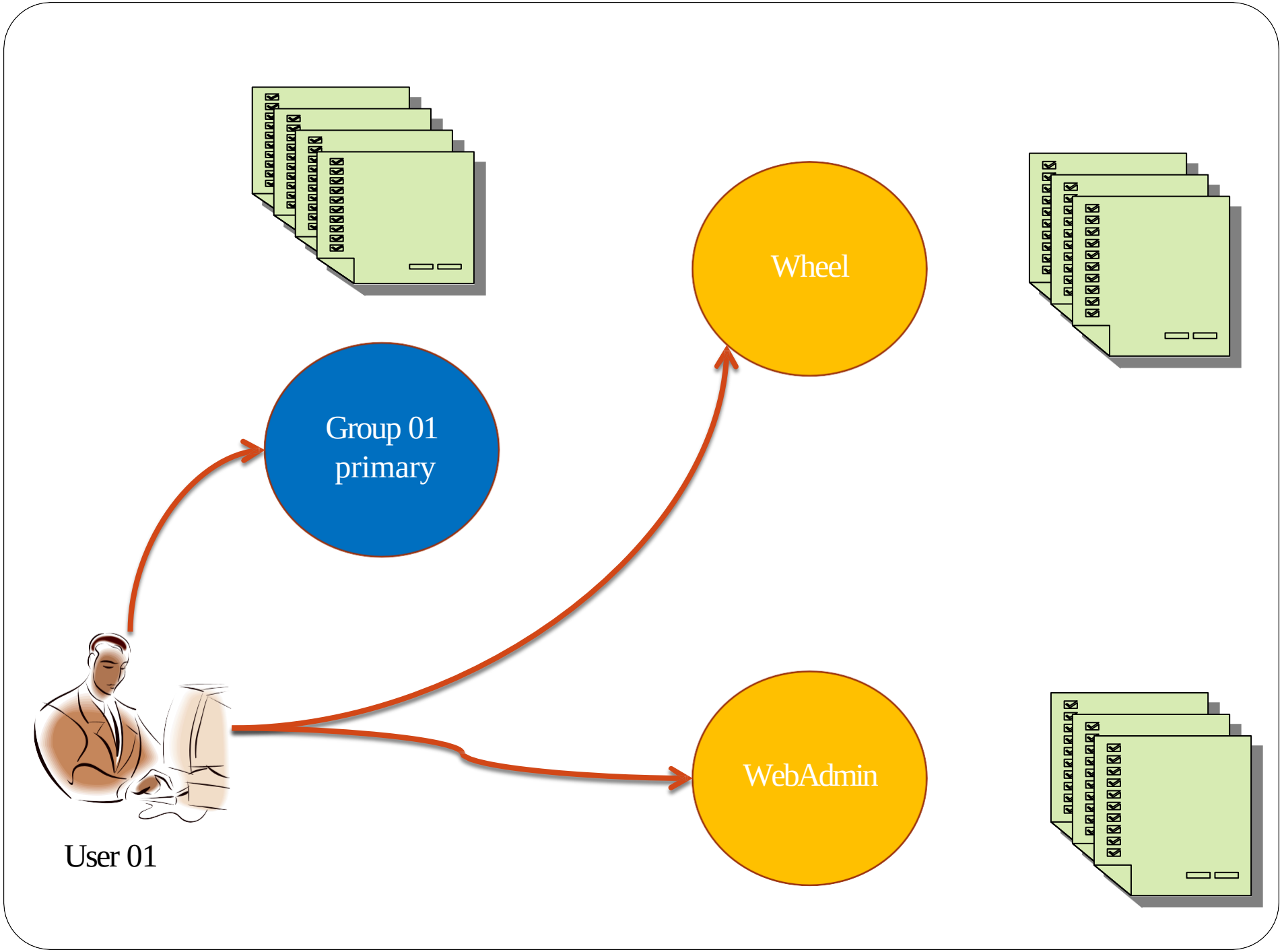
Primary Groups and Supplementary Groups

- Every user has exactly one primary grouplisted by GID number in the /etc/passwd file.

- When you create a new regular user, a new group with the same name as that user is created.This group is used as the primary group for the new user, and that user is theonly member of this *User Private Group*

Supplementary Groups

- Membership in supplementary groups is determined by the **/etc/group** file
- Users are granted access to files based on whether any of their groups have access.


Wheel
Group 01
primary
WebAdmin
User 01

GAINING SUPERUSER ACCESS

THE SUPERUSER

- A user that has all power over the system.
- This user has the power to:
  - override normal privileges on the file system,
  - manage and administer the system.
  - perform tasks such as installing or removing software
  - manage system files and directories
  - Manage "fixed" hard drives

But

- He has unlimited power to damage the system.

➔ Administrators should log in to the system as an unprivileged user and use various tools to temporarily gain root privileges.

SWITCHING USERS

- **su** command allows users to switch to a different user account.
- If you run su from a regular user account, you will be prompted for the password of the account to which you want to switch.

```
[user01@host ~]$ su - user02
Password:
[user02@host ~]$
```

If you omit the user name, the **su** or **su -** command attempts to switch to **root** by default.

```
[user01@host ~]$ su -
Password:
[root@host ~]#
```

RUNNING COMMANDS WITH SUDO

- **sudo** requires users to enter their own password for authentication, not the password of the user account they are trying to access
- **sudo** can be configured to allow specific users to run any command (or some specific) as some other user.

```
[user01@host ~]$ sudo usermod -L user02
[sudo] password for user01:
[user01@host ~]$ su - user02
Password:
su: Authentication failure
[user01@host ~]$
```

- all commands executed are logged by default to /var/log/secure.

```
[user02@host ~]$ sudo tail /var/log/secure
[sudo] password for user02:
user02 is not in the sudoers file.  This incident will be reported.
[user02@host ~]$
```

```
[user01@host ~]$ sudo tail /var/log/secure
...output omitted...
Feb  6 20:45:46 host sudo[2577]:  user01 : TTY=pts/0 ; PWD=/home/user01 ;
 USER=root ; COMMAND=/sbin/usermod -L user02
...output omitted...
```

MANAGING LOCAL USER ACCOUNTS

Creating Users

1. **useradd *username*:** creates a new user named username.
2. It sets up the user's home directory and account information,
3. It creates a private group for the user named username.
4. At this point the account does not have a valid password set, and the user cannot login until a password is set.

Modifying Existing Users

usermod –*option value* username

USERMOD OPTIONS:	USAGE
-c, --comment COMMENT	Add the user's real name to the comment field.
-g, --gid GROUP	Specify the primary group for the user account.
-G, --groups GROUPS	Specify a comma-separated list of supplementary groups for the user account.
-a, --append	Used with the **-G** option to add the supplementary groups to the user's current set of group memberships instead of replacing the set of supplementary groups with a new set.
-d, --home HOME_DIR	Specify a particular home directory for the user account.
-m, --move-home	Move the user's home directory to a new location. Must be used with the **-d** option.
-s, --shell SHELL	Specify a particular login shell for the user account.
-L, --lock	Lock the user account.
-U, --unlock	Unlock the user account.

Deleting Users

1. **userdel *username*:** removes the details of username from /etc/passwd, but leaves the user's home directory intact.

2. **userdel -r *username*** removes the details of username from /etc/passwd and also deletes the user's home directory.

Users Passwords

- **passwd *username*** command sets the initial password or changes the existing password of username.
- The root user can set a password to any value.

```
[root@host ~]# passwd user01
Changing password for user user01.
New password: redhat
BAD PASSWORD: The password fails the dictionary check - it is based on a
 dictionary word
Retype new password: redhat
passwd: all authentication tokens updated successfully.
[root@host ~]#
```

RESTRICTING ACCESS

- **usermod** command can lock an account with the **-L** option.

```
[user01@host ~]$ sudo usermod -L user03
[user01@host ~]$ su - user03
Password: redhat
su: Authentication failure
```

UID Ranges

- *UID 0* is always assigned to the superuser account, **`root`**.
- *UID 1-200* is a range of "system users" assigned statically to system processes
- *UID 201-999* is a range of "system users" used by system processes that do not own files on the file system. They are typically assigned dynamically from the available pool when the software that needs them is installed. Programs run as these "unprivileged" system users in order to limit their access to only the resources they need to function.
- *UID 1000+* is the range available for assignment to regular users.

MANAGING LOCAL GROUP ACCOUNTS

Creating Groups

- **groupadd *groupname***
- Without options it uses the next available GID from the range specified in the /etc/login.defs
- The **-g** option specifies a particular GID for the group to use.

```
[user01@host ~]$ sudo groupadd -g 10000 group01
[user01@host ~]$ tail /etc/group
...output omitted...
group01:x:10000:
```

- The -r option creates a system group using a GID from the range of valid system GIDs listed inthe /etc/login.defs

```
[user01@host ~]$ sudo groupadd -r group02
[user01@host ~]$ tail /etc/group
...output omitted...
group01:x:10000:
group02:x:988:
```

Modifying Existing Groups

- **groupmod** :changes the properties of an existing group.

```
[user01@host ~]$ sudo groupmod -n group0022 group02
[user01@host ~]$ tail /etc/group
 ...output omitted...
 group0022:x:988:
```

```
[user01@host ~]$ sudo groupmod -g 20000 group0022
[user01@host ~]$ tail /etc/group
...output omitted...
group0022:x:20000:
```

Deleting Groups

- **groupdel** command removes groups
- We cannot remove a group if it is the primary group of any existing user.
- check all file systems to ensure that no files remain on the system that are owned by the group

```
[user01@host ~]$ sudo groupdel group0022
```

Changing Group Membership

- The membership of a group is controlled with user management. Use the **usermod -g** command to change a user's primary group.

```
[user01@host ~]$ id user02
uid=1006(user02) gid=1008(user02) groups=1008(user02)
[user01@host ~]$ sudo usermod -g group01 user02
[user01@host ~]$ id user02
uid=1006(user02) gid=10000(group01) groups=10000(group01)
```

- Use the **usermod -aG** command to add a user to a supplementary group.

```
[user01@host ~]$ id user03
uid=1007(user03) gid=1009(user03) groups=1009(user03)
[user01@host ~]$ sudo usermod -aG group01 user03
[user01@host ~]$ id user03
uid=1007(user03) gid=1009(user03) groups=1009(user03),10000(group01)
```